

Reverse Zone File

Created:

```
/etc/bind/db.10.1.10
```

Contents:

```
$TTL 604800

@ IN SOA ns1.worldskills.ph. admin.worldskills.ph. (
    2026052901
    604800
    86400
    2419200
    604800 )

@ IN NS ns1.worldskills.ph.

10 IN PTR int-srv01.worldskills.ph.
```

Understanding PTR Records

PTR records are used for reverse DNS lookups.

Instead of translating:

```
Hostname → IP Address
```

PTR records translate:

```
IP Address → Hostname
```

Example:

```
10.1.10.10
```

↓

```
int-srv01.worldskills.ph
```

Configuration Validation

Before restarting BIND, configuration files were validated.

Validate Main Configuration

Command:

```
named-checkconf
```

Expected:

```
(no output)
```

Result:

```
(no output)
```

This indicates the configuration syntax is valid.

Validate Forward Zone

Command:

```
named-checkzone worldskills.ph /etc/bind/db.worldskills.ph
```

Output:

```
zone worldskills.ph/IN:  
loaded serial 2026052901
```

OK

Validate Reverse Zone

Command:

```
named-checkzone 10.1.10.in-addr.arpa /etc/bind/db.10.1.10
```

Output:

```
zone 10.1.10.in-addr.arpa/IN:  
loaded serial 2026052901
```

OK

Reload DNS Service

Command:

```
systemctl reload named
```

Verification:

```
systemctl status named
```

Output:

```
active (running)
```

Forward Lookup Testing

Testing from the DNS server itself.

Test Name Server Record

Command:

```
dig @10.1.10.10 ns1.worldskills.ph
```

Output:

```
ANSWER SECTION:  
  
ns1.worldskills.ph.  
604800 IN A 10.1.10.10
```

PASS

Test Server Record

Command:

```
dig @10.1.10.10 int-srv01.worldskills.ph
```

Output:

```
ANSWER SECTION:  
  
int-srv01.worldskills.ph.  
604800 IN A 10.1.10.10
```

PASS

Reverse Lookup Failure

Initial test:

```
dig @10.1.10.10 -x 10.1.10.10
```

Unexpected output:

```
status: NXDOMAIN
```

Reverse DNS was failing.

Forward DNS worked.

Reverse DNS did not.

Investigation

Checked reverse zone file.

Command:

```
cat /etc/bind/db.10.1.10
```

Observed:

```
10 IN PTR ns1.worldskills.ph.  
10 IN PTR int-srv01.worldskills.ph.
```

Two PTR records existed for the same host.

Root Cause Analysis

The reverse zone definition was technically valid.

However:

```
One IP  
Two PTR records
```

created ambiguity.

Although BIND loaded the zone successfully, reverse lookups did not return the expected result.

Resolution

Modified:

```
10 IN PTR int-srv01.worldskills.ph.
```

Removed:

```
10 IN PTR ns1.worldskills.ph.
```

Reloaded:

```
systemctl reload named
```

Reverse Lookup Retest

Command:

```
dig @10.1.10.10 -x 10.1.10.10
```

Output:

```
ANSWER SECTION:
```

```
10.10.1.10.in-addr.arpa.
```

```
PTR int-srv01.worldskills.ph.
```

PASS

Additional Verification

Command:

```
host 10.1.10.10
```

Output:

```
10.1.10.10.in-addr.arpa
domain name pointer
int-srv01.worldskills.ph.
```

PASS

OPNsense DHCP Integration

At this stage:

DNS worked.

The next goal was distributing DNS settings automatically through DHCP.

DHCP Scope

Network:

```
10.1.10.0/24
```

Gateway:

```
10.1.10.1
```

DNS:

```
10.1.10.10
```

Pool:

```
10.1.10.100 - 10.1.10.200
```

Kea DHCP Configuration

Configured:

Services
→ Kea DHCPv4

Subnet:

10.1.10.0/24

Description:

INT

Pool:

10.1.10.100 - 10.1.10.200

DHCP Options Configured

Gateway:

10.1.10.1

DNS Server:

10.1.10.10

Domain Name:

worldskills.ph

Domain Search:

worldskills.ph

NTP Server:

10.1.10.10

Time Server:

10.1.10.10

Expected Client Behavior

After receiving a DHCP lease, clients should obtain:

IP Address

10.1.10.x

Gateway

10.1.10.1

DNS

10.1.10.10

Search Domain

worldskills.ph

Ubuntu Client Testing

Client received:

10.1.10.175

Gateway:

```
10.1.10.1
```

Verified:

```
nmcli dev show
```

Output:

```
IP4.ADDRESS:  
10.1.10.175/24  
  
IP4.GATEWAY:  
10.1.10.1
```

Initial DNS Failure

Testing:

```
nslookup int-srv01.worldskills.ph
```

Output:

```
NXDOMAIN
```

Unexpected.

DNS server itself worked.

Client could not resolve names.

Direct DNS Test

Command:

```
nslookup int-srv01.worldskills.ph 10.1.10.10
```

Output:

```
Name:
int-srv01.worldskills.ph

Address:
10.1.10.10
```

PASS

This proved:

```
BIND is working
Client connectivity is working
DNS queries reach server
```

Problem was elsewhere.

DHCP Investigation

Checked active DNS configuration.

Command:

```
resolvectl dns
```

Output:

```
10.1.10.1
```

Unexpected.

Expected:

```
10.1.10.10
```

Search Domain Investigation

Command:

```
resolvectl domain
```

Output:

```
internal
```

Unexpected.

Expected:

```
worldskills.ph
```

This indicated DHCP was still distributing incorrect values.

Lease Analysis

Command:

```
sudo grep -E  
"domain-name|domain-name-servers"  
/var/lib/dhcp/dhclient*.leases
```

Output:

```
option domain-name-servers 10.1.10.1;  
  
option domain-name "worldskills.ph";
```

Critical finding:

Clients were still receiving:

```
DNS = 10.1.10.1
```

instead of:

```
DNS = 10.1.10.10
```

Discovery of DHCP Conflict

Investigation inside OPNsense revealed:

Two DHCP-capable services existed:

```
Dnsmasq DNS & DHCP
```

and

```
Kea DHCP
```

Both could provide DHCP-related options.

This created conflicting DHCP information.

Root Cause

Dnsmasq was still advertising:

```
DNS Server = 10.1.10.1  
Domain = internal
```

while Kea DHCP advertised:

```
DNS Server = 10.1.10.10  
Domain = worldskills.ph
```

Clients received inconsistent DHCP settings.

Resolution

Disabled:

Services
→ Dnsmasq DNS & DHCP

Disabled:

DNS

Disabled:

DHCP

Retained:

Kea DHCP

as the only DHCP service.